Exercise 2B: Poisson regression

For this exercise we are going to use a data set which can be found here:
`https://stats.idre.ucla.edu/stat/data/poisson_sim.csv`
The data set contains three relevant variables:

- `num_awards`, a counting variable representing the number of awards won in high school by a student;
- `prog`, a categorical factor variable representing the type of program the student was enrolled in: 1=general, 2=academic and 3=vocational; and
- `math`, a continuous variable representing the student's final exam math score.

We want to predict `num_awards` based on `prog` and `math`.

- Plot the data with `num_awards` on the y-axis and the points coloured differently for different values of `prog`

As we can see from the plot, a linear regression is not appropriate here. Since we are dealing with counting data we will instead use a Poisson regression, assuming

$$y_{i,j} \sim \text{Pois}(\exp(\alpha_j + \beta x_i))$$

where j encodes the value of prog, x_i is the math score, and $y_{i,j}$ is the number of awards. In words, what we assume is that the number of awards given the maths score and the type of program is Poisson distributed, where the mean of that distribution is given by the regression model $\exp(\alpha_j + \beta x_i)$. That is, we need to estimate α_1, α_2, α_3 and β. The log likelihood function will be constructed as follows: Let

$$f(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!},$$

i.e $f_\lambda(x)$ is a Poisson pdf, then the log likelihood function is

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \sum_{j=1}^{3} \log f(y_{i,j}; \exp(\alpha_j + \beta x_i))$$

- Implement the log likelihood function.
- Maximise the log likelihood to find the parameters α_1, α_2, α_3 and β using a numerical optimiser, for example `minimize()` from `scipy.optimize` (remember that maximising $\ell(x)$ is the same as minimising $-\ell(x)$).
- Draw in the three (one for each student program) means of $y_{i,j}$ (i.e draw $\exp(\alpha_j + \beta x_i)$) in the scatter plot you made in the first exercise.
- How do you interpret the α_j coefficient?
- *(optional)* Estimate the same model with the `statsmodels` package.